

Loss Term	Formulation	Soft vs. Hard	Type	Comments
Schrödinger residual (physics loss)	$\mathcal{L}_{\text{TISE}} = \sum_n \left\ -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi_n^\theta + V_\theta(x) \psi_n^\theta(x) - E_n^\theta \psi_n^\theta(x) \right\ ^2$	soft	Consistency condition	Enforces the TISE.
Wavefunction normalization loss	$\mathcal{L}_{\text{norm}} = \frac{1}{N} \sum_{n=1}^N \left(\int \ \psi_n^\theta(x)\ ^2 dx - 1 \right)^2$	soft	Regularizer	Enforces normalization structure.
Scale-aware smoothness	$\mathcal{L}_{\text{smooth}} = \left\langle \frac{\ V_\theta''(x)\ ^2}{\epsilon + \ V_\theta(x)\ ^2} \right\rangle$	soft	Regularizer	Penalizes steep curvature in $V_\theta(x)$. Encourages physically plausible potentials and stabilizes the inverse problem.
Data mismatch (observables)	$\mathcal{L}_{\text{data}} = \frac{1}{N_s} \sum_n (E_n^\theta - E_n^{\text{obs}})^2 + \frac{1}{N_s} \sum_n \frac{1}{M} \sum_j \left(\psi_n^\theta(x_j) ^2 - \rho_n^{\text{obs}}(x_j) \right)^2$	soft (measurement noise)	Consistency condition	Ensures learned eigenstates match noisy observations. Note that N_s denotes the number of eigenstates learned and M denotes the number of 1-D spatial grid points used as colloaction points.
Total Loss	$\mathcal{L}_{\text{total}} = \lambda_{\text{TISE}} \mathcal{L}_{\text{TISE}} + \lambda_{\text{norm}} \mathcal{L}_{\text{norm}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}} + \lambda_{\text{data}} \mathcal{L}_{\text{data}}$	soft	Composite objective	Weights are static scalars.